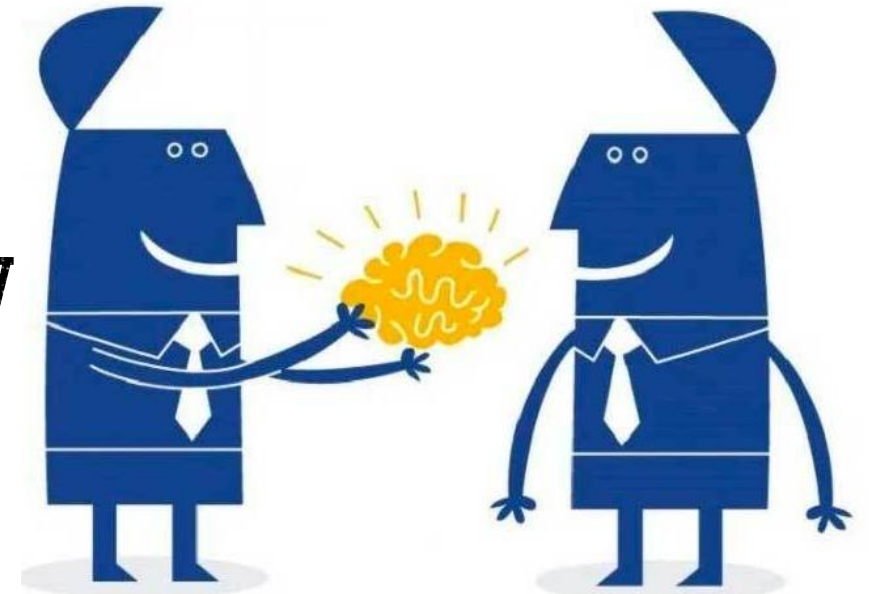


Session 9: Applying Transfer Learning for Cost-Efficiency

Presented by: Ahmad Bany Abdelnabi

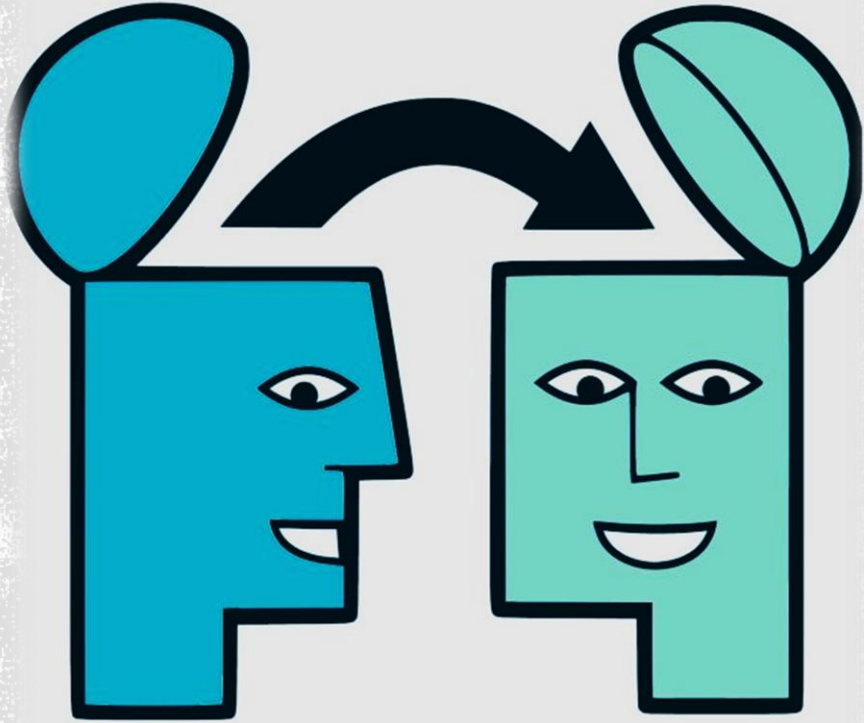
School of Modeling, Simulation, and Training, University of Central Florida



*This activity
is supported by:*

The NATO Science for Peace
and Security Programme

Agenda Overview



Recap

PART 1: Foundation

- Transfer Learning Fundamentals
- Segmentation (U-Net) & Detection (YOLO) Architectures
- Cost-Efficiency Strategies

PART 2: Implementation

- Environment Setup
- Segmentation Task: U-Net for flood/debris mapping
- Detection Task: YOLO for survivor detection
- Optimization Techniques

PART 3: Discussions & Wrap-up

- Discussions
- Q&A

Figures adapted from different resources for illustration only. References will be included in the last slide.



WHAT DID WE LEARN SO FAR?

Session 1: Introduction to AI for data analytics and aerial imaging

- Fundamentals of using AI/ML for data analytics and aerial imaging

Session 2: Understanding Image Classification and CNNs

- Foundational models for image analysis.
- Core components of CNNs.

Session 3: The Building Blocks of CNNs

- Continued the exploration of CNNs

Session 4: Effective Training of Deep Learning Models

- Strategies for training models effectively.

Session 5: End-to-End Workflow for Deep Learning

- A practical, step-by-step walkthrough of a complete DL project.



WHAT DID WE LEARN SO FAR?

Session 6: Case Study on Natural Hazard Detection

- A real-world scenario where CNNs are used to identify building damage



Challenges and Opportunities for AI Adoption in HL



Session 7: ML Workflows for Disaster Management Solutions

- entire data pipeline, data quality, and model deployment considerations for real-time systems



Integrating remote sensing and GeoAI

- remote sensing data with GeoAI for large-scale disaster management.



Session 8: Introduction to Automated Building Damage and Flood Event Detection

challenges & techniques for detecting building damage and flood

AI, ML, & DL (REFRESHER)

- **Artificial Intelligence (AI)**

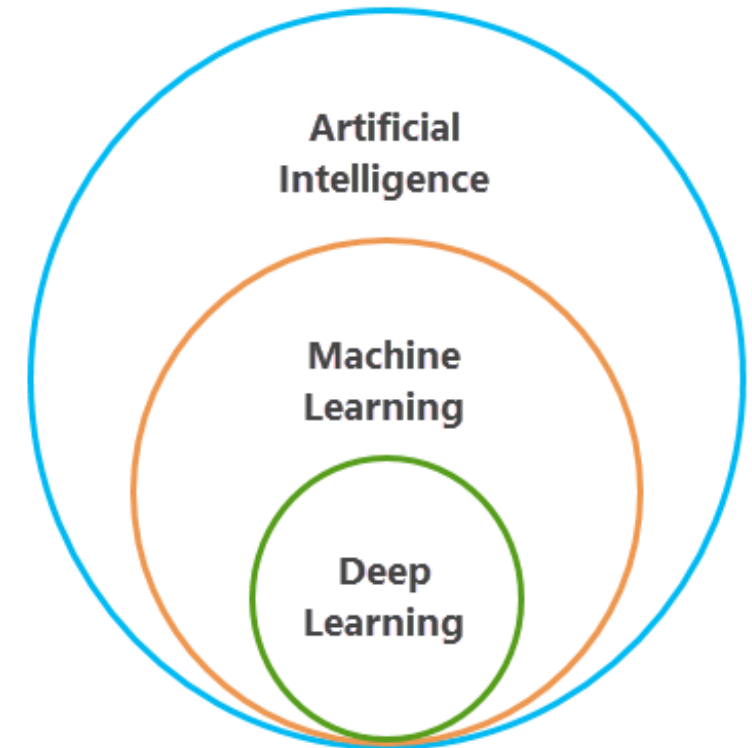
- The broadest concept shown in the outer circle.
- Refers to **systems or machines that mimic human intelligence** to perform tasks like reasoning, learning, and problem-solving.

- **Machine Learning (ML)**

- A subset of AI represented by the middle circle.
- Focuses on **algorithms that allow systems to learn from data** and improve performance without explicit programming.

- **Deep Learning (DL)**

- A subset of Machine Learning represented by the innermost circle.
- Uses **neural networks with multiple layers to analyze complex patterns in large datasets**, often applied in image recognition, natural language processing, etc.





DEEP LEARNING IN DISASTER RESPONSE – APPLICATIONS



OPTiSiM
Optimization & Simulation Lab

Human Detection & Localization

- Identify survivors in aerial or satellite imagery

Damage Assessment

- Classify and quantify infrastructure damage from drone or satellite images.

Resource Allocation

- Predict demand for medical supplies and food.

Disaster Prediction & Early Warning

- Use RNNs/LSTMs for forecasting floods, hurricanes, or wildfires.

Communication Restoration

- Optimize UAV-based networks for connectivity in affected areas.



DEEP LEARNING IN DISASTER RESPONSE – APPROACHES

Convolutional Neural Networks (CNNs)

- For image-based tasks like detection and damage classification.

Recurrent Neural Networks (RNNs) & LSTMs

- For sequential data such as weather patterns and sensor readings.

Generative Models (GANs)

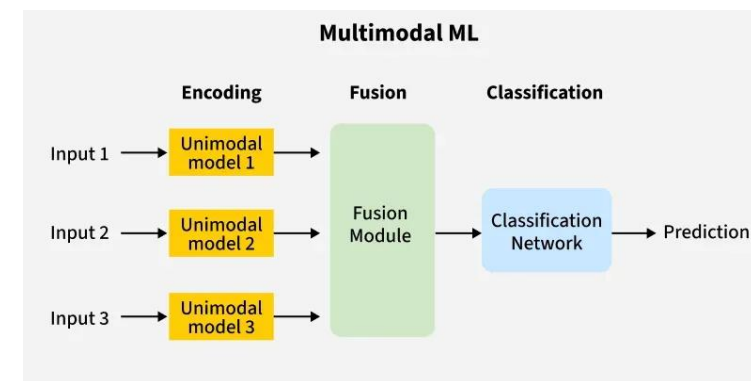
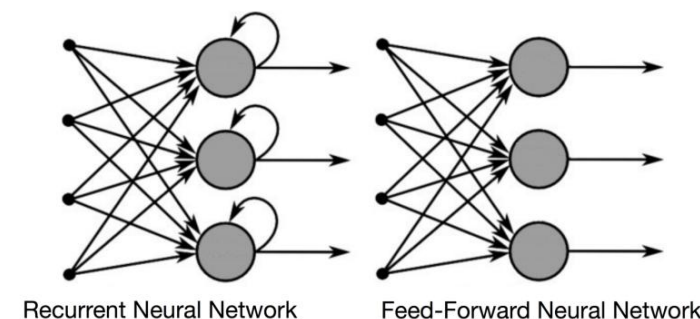
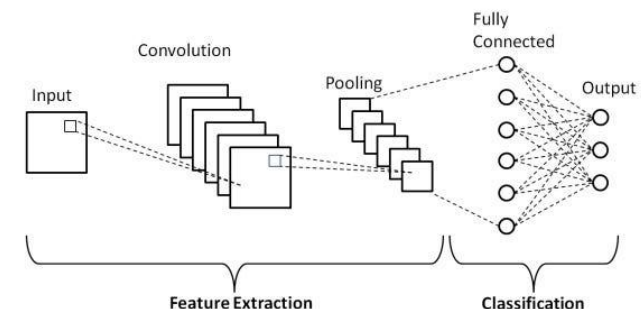
- For simulating disaster scenarios and augmenting training datasets.

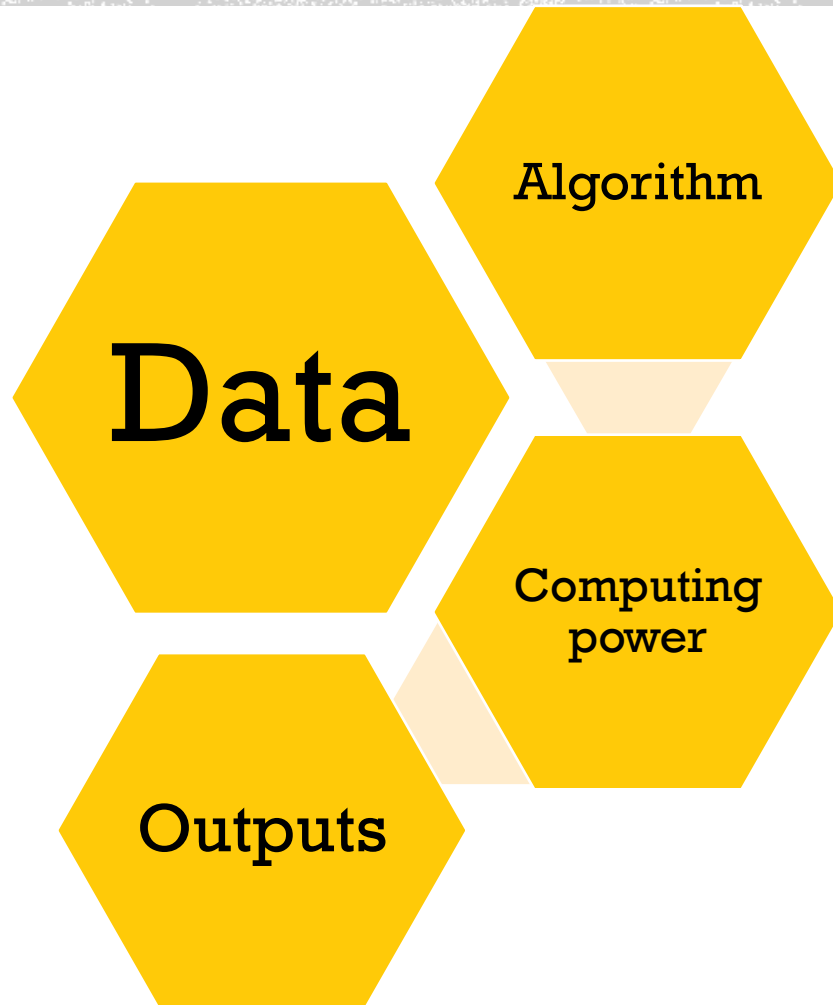
Transfer Learning

- Adapt pre-trained models for disaster-specific datasets.

Multimodal Learning

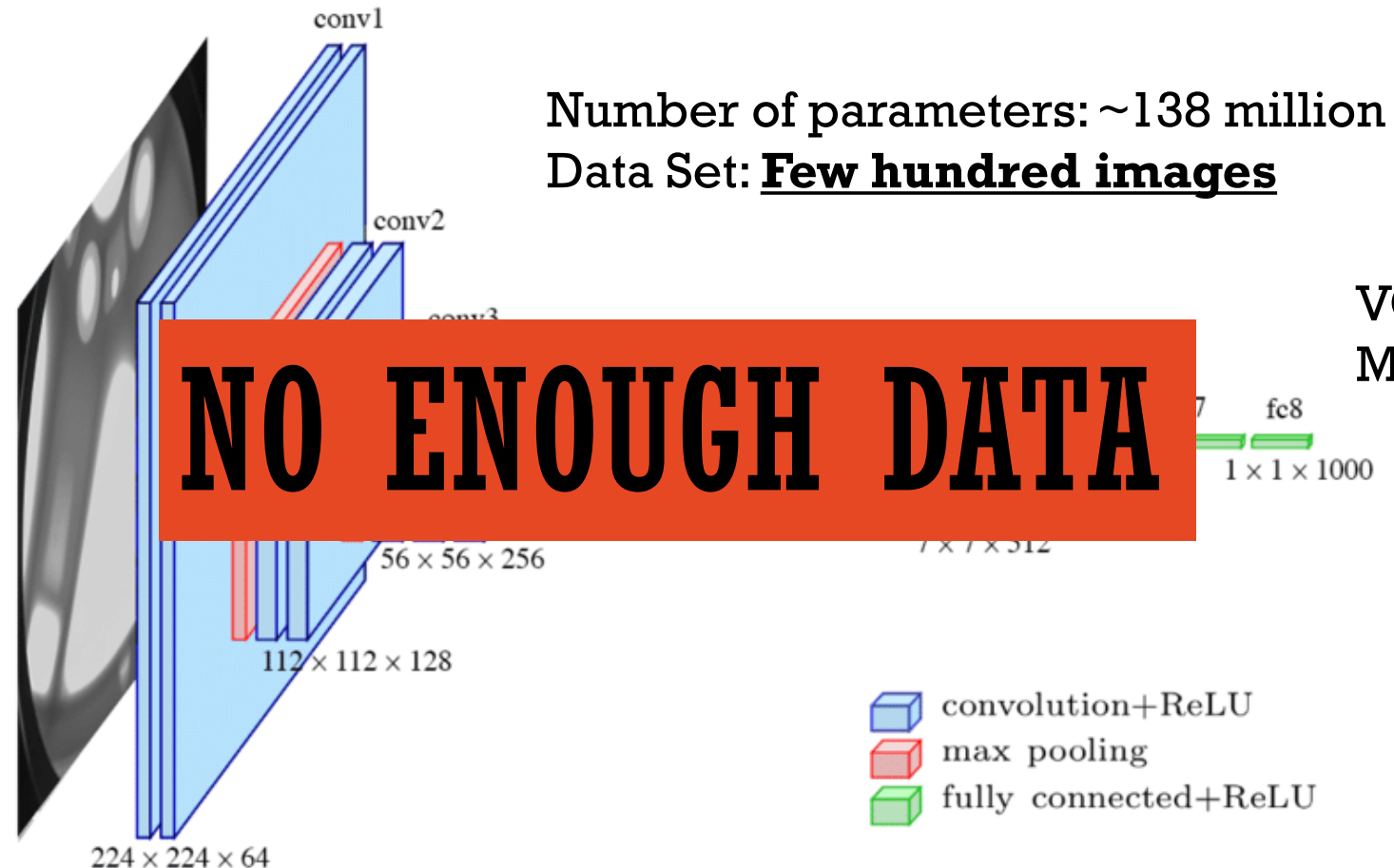
- Combine satellite imagery, social media, and sensor data for holistic analysis.





**COMPONENTS
OF ANY AI
MODEL?**

MOTIVATION

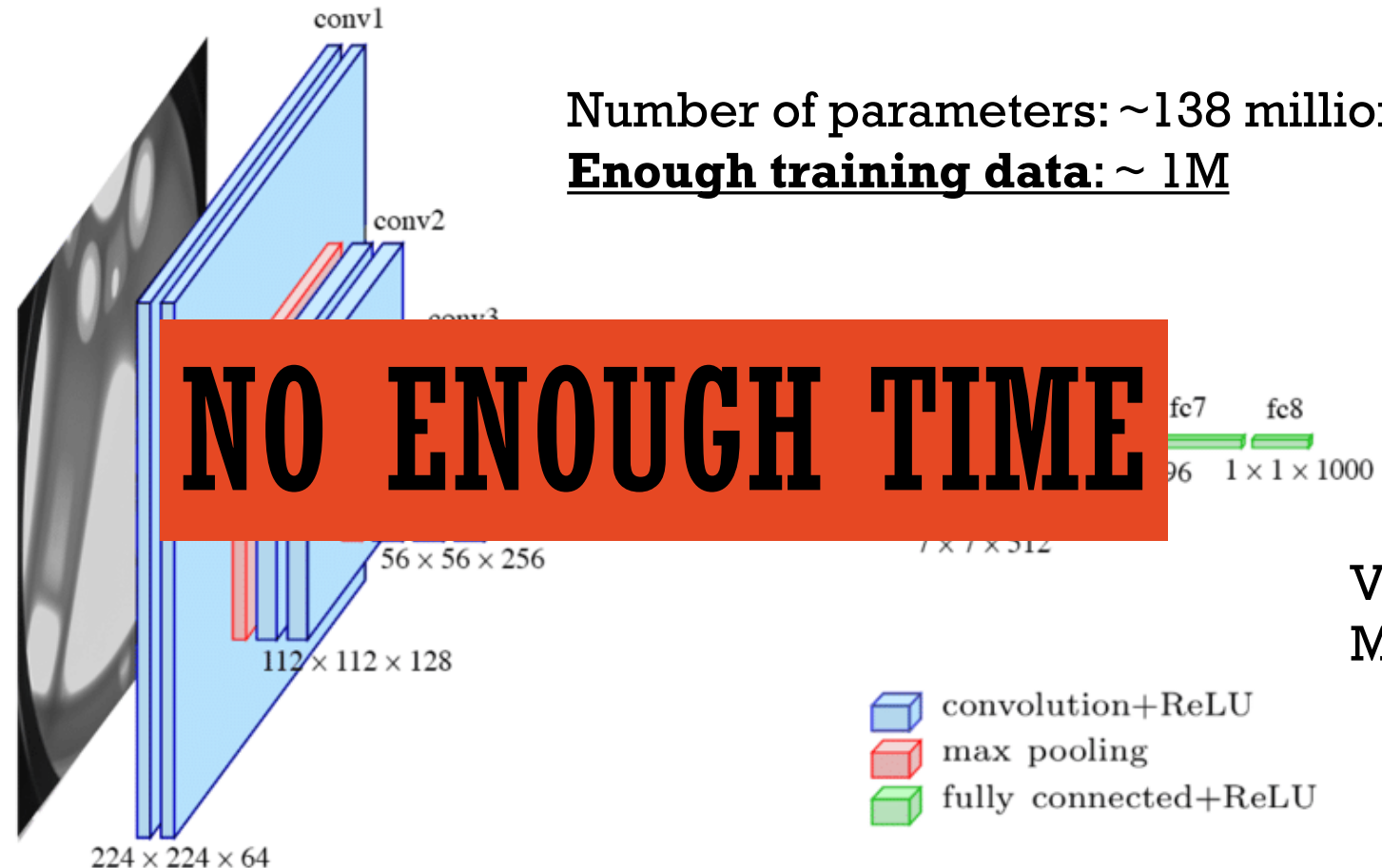


Number of parameters: ~138 million
Data Set: **Few hundred images**

VGG16
Model

If a model like VGG16 has 138 million parameters, and we only have a few hundred disaster images, **do you think we can train it from scratch?**

MOTIVATION



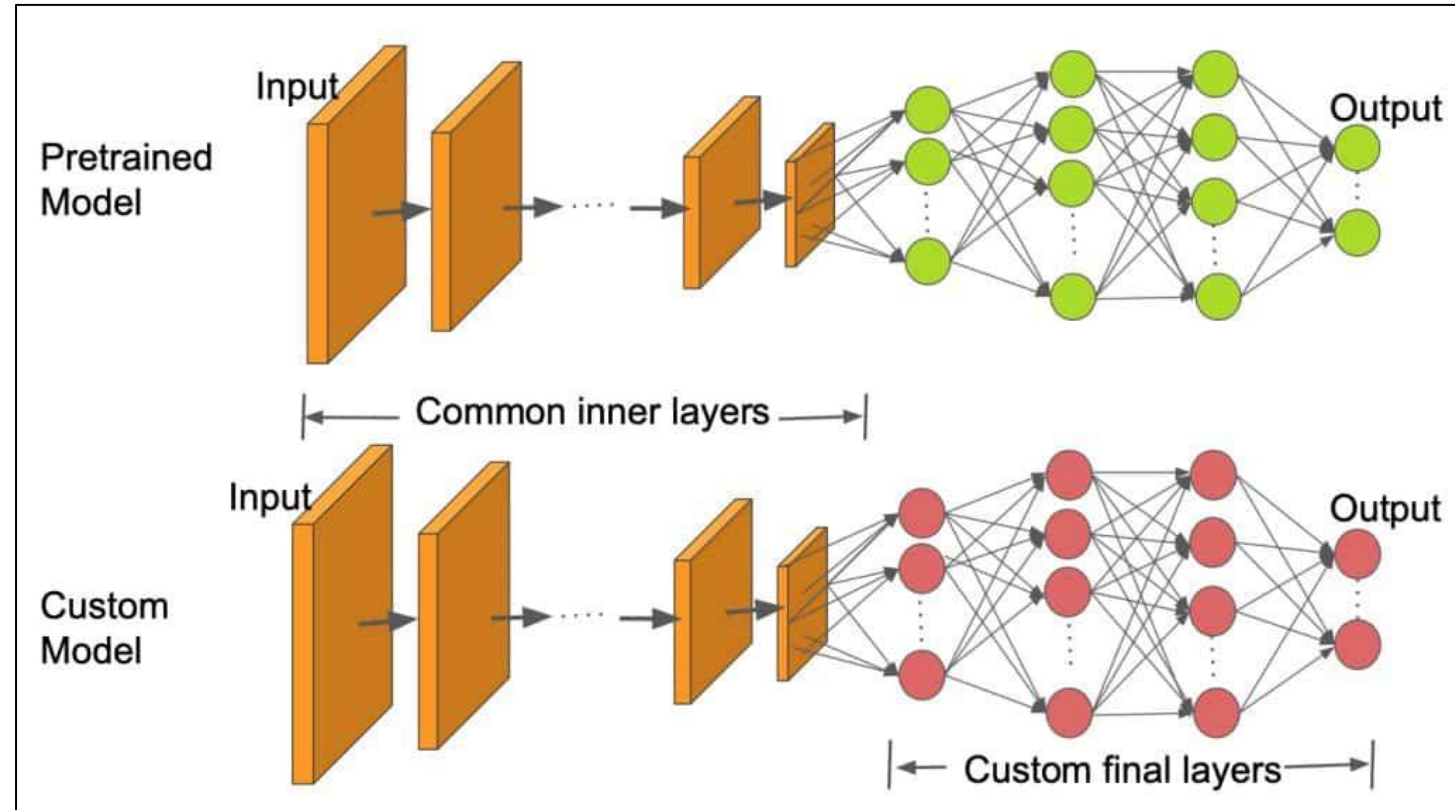
Number of parameters: ~138 million
Enough training data: ~ 1M

VGG16
Model

Imagine a hurricane just hit—how much time do we have to collect and label data and to train and deploy an AI model?

BASIC IDEA OF TRANSFER LEARNING

- **Definition:** Adapting pre-trained models for new tasks
- **Why It Matters:**
 - Limited labeled disaster data
 - Faster deployment during emergencies
- **Approaches (for this lecture):**
 - **Segmentation:** Fine-tune U-Net for flood or debris mapping
 - **Detection:** Fine-tune YOLO for locating survivors, vehicles, hazards
- **Benefits:**
 - Reduces training time
 - Improves accuracy with small datasets



BASIC IDEA OF TRANSFER LEARNING

■ Types of TL

Train a ML model M for a task T using a dataset D_s

Use model M on a **new dataset** D_a for the same task T

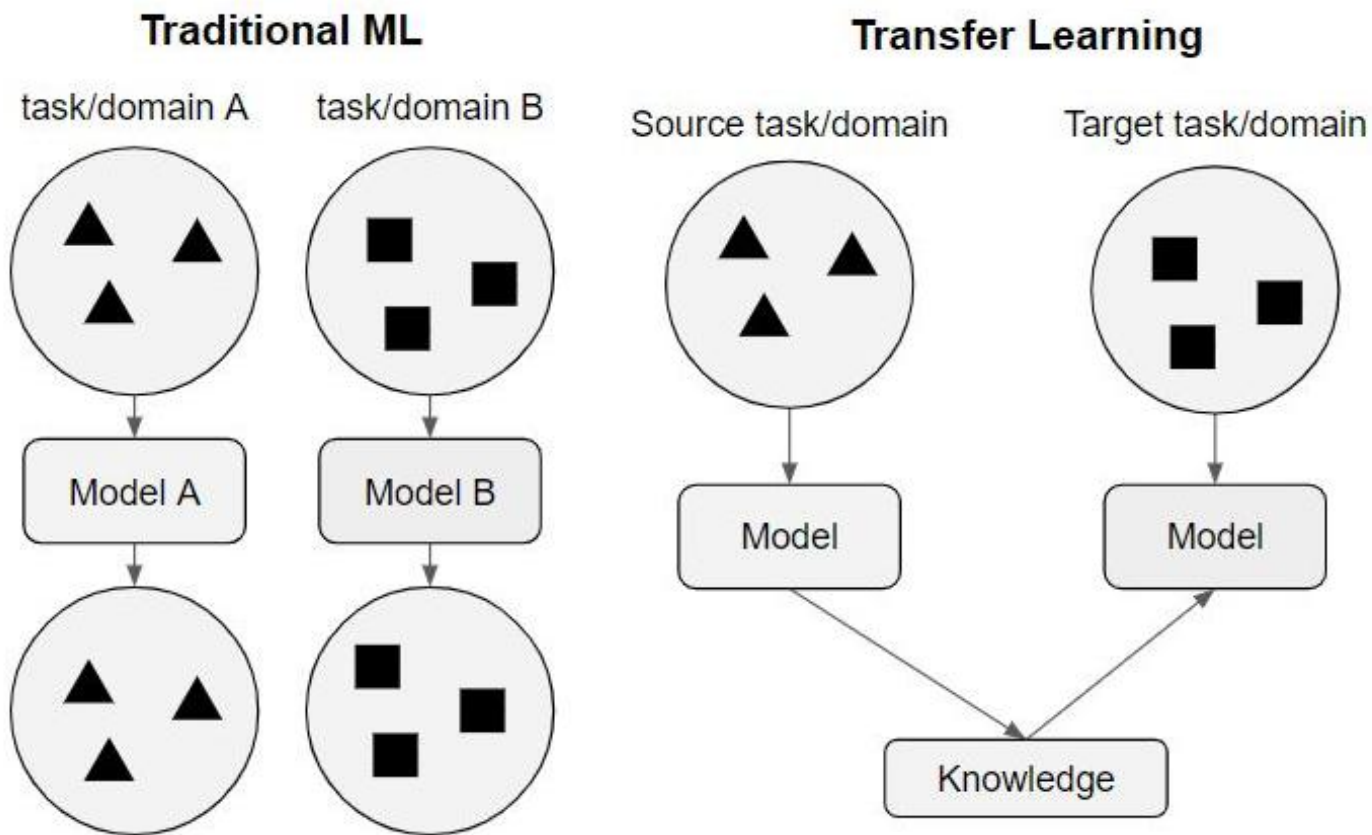
Use part of M on original dataset D_s for a **new task** T_n

Use part of M on a **new dataset** D_a for a **new task** T_n

TL is a research problem in ML that focuses on **storing knowledge** gained while solving one problem and applying it to a different but related problem.

BASIC IDEA OF TRANSFER LEARNING

Training and evaluation on the same task or domain.



Storing knowledge gained solving one problem and applying it to a different but related problem.

THE ANALOGY: Learning to drive a car → Learning to drive a truck

- You **do not relearn** steering, braking, traffic rules
- You **adapt existing knowledge** to new vehicle characteristics



DEEP LEARNING VS TRANSFER LEARNING

	TRADITIONAL TRAINING (From Scratch)	TRANSFER LEARNING
Data Requirements	100K+ labeled images	1K - 10K labeled images (90% reduction)
Training Time	Days to weeks	Hours to 1-2 days (95% reduction)
Compute Cost	\$5,000 - \$50,000+	\$50 - \$500 (90-95% reduction)
Accuracy	High (if sufficient data)	Comparable or better
Time to Deployment	Weeks to months	Days
Expertise Required	Deep ML expertise	Moderate ML knowledge
Risk	High (model may not converge)	Low (starting from proven model)

Transfer learning leverages billions of parameters already learned from large datasets (ImageNet, COCO, etc.)



WHAT NEURAL NETWORKS LEARN: LAYER BY LAYER

- **Early Layers: UNIVERSAL FEATURES** → Transfer rate: ~95% (almost always useful)
 - Edge detectors (horizontal, vertical, diagonal)
 - Color blobs (red, green, blue regions)
 - Simple textures
- **Middle Layers: SEMI-UNIVERSAL FEATURES** → Transfer rate: ~80% (usually useful)
 - Corners and contours
 - Textures (wood, metal, water, fabric)
 - Simple patterns (stripes, grids, dots)
 - Basic shapes (circles, rectangles)
- **Deep Layers: DOMAIN-SPECIFIC FEATURES** → Transfer rate: ~50% (sometimes useful)
 - Object parts (wheels, windows, faces)
 - Complex textures and patterns
 - Pose and orientation information
 - Object templates
- **Final Layers: TASK-SPECIFIC FEATURES** → Transfer rate: ~10% (rarely useful—usually retrained)
 - Class-specific detectors ("cat," "car," "airplane")
 - Task-specific decision boundaries
 - Output layer (tailored to original task)

STRATEGY: Keep universal features, adapt specific features

- For similar tasks: Fine-tune only final layers
- For different tasks: Fine-tune more layers

- **How do you build an image classifier for disaster scenarios with limited data and compute resources?**
- **Use Pre-trained Models**
 - Start with models trained on large datasets (e.g., ImageNet) to leverage learned features.
- **Main Idea:**
 - Early layers capture **generic visual features** (edges, textures, shapes).
 - Later layers adapt to **domain-specific features** (e.g., collapsed buildings, flooded areas, injured individuals).
 - Fine-tune only the last few layers and fully connected layers for disaster-specific tasks.

Take a CNN pre-trained on ImageNet.



Retrain it on a small dataset of drone or satellite images to classify:

Human presence for search and rescue	Infrastructure damage levels for resource allocation.	Flooded vs. safe zones for evacuation planning.
--------------------------------------	---	---



TRANSFER LEARNING STRATEGIES

STRATEGY	1: FEATURE EXTRACTION (Frozen Backbone)	2: PARTIAL FINE-TUNING (Selective Unfreezing)	3: FULL FINE-TUNING (All Layers)	4: PROGRESSIVE UNFREEZING
# Frozen Layers	All encoder layers	Early layers	None	Gradually unfreeze layers
# Trainable Layers	Only task head	Deep layers + head	All layers (low learning rate)	Expands
Cost	Lowest	Low-Medium	Medium	Medium
Time	Fastest	Fast	Moderate	Moderate
Accuracy	Good	Better	Best	Excellent
Best for	<1K images, very similar domains, rapid prototyping	1K-5K images, moderately similar domains	5K+ images, different domains, max performance	Complex domains, preventing catastrophic forgetting

PRE-TRAINED MODELS AS FEATURE EXTRACTORS

INPUT IMAGE (Aerial disaster scene)

FROZEN ENCODER - Pre-trained weights

- Convolutional layers extract features
- Pooling layers reduce dimensionality
- Batch normalization stabilizes training

HIGH-DIMENSIONAL FEATURE VECTOR (e.g., 2048 dimensions capturing image essence)

TRAINABLE TASK HEAD

- Fully connected layers for classification
- OR Decoder for segmentation
- OR Detection head for bounding boxes

TASK-SPECIFIC OUTPUT

WHEN TO USE:

- Very limited training data (<1K images)
- Extremely tight computational budget
- Rapid prototyping and iteration
- Source and target domains are very similar

WHEN TRANSFER LEARNING WORKS (AND WHEN IT DOESN'T)

TRANSFER LEARNING EXCELS WHEN:

1. DOMAIN SIMILARITY

- Source: Natural outdoor images (ImageNet)
- Target: Aerial disaster images
- Reason: Shared visual features (buildings, terrain)

2. TASK SIMILARITY

- Source: Object detection (COCO)
- Target: Vehicle/survivor detection
- Reason: Both require localization and classification

3. SUFFICIENT (BUT LIMITED) TARGET DATA

- Sweet spot: 500 - 10,000 labeled images
- Too few (<100): transfer struggles
- Too many (>100K): Training from scratch competitive

4. FEATURE REUSABILITY

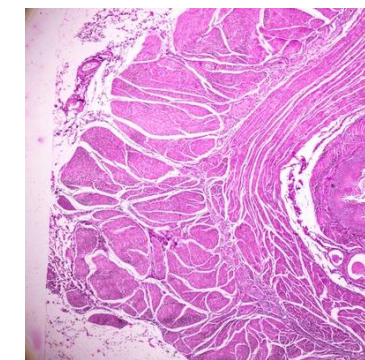
- Low-level features (edges, textures): Always transfer
- High-level features (object parts): Transfer if domain is similar

WHEN TRANSFER LEARNING WORKS (AND WHEN IT DOESN'T)

TRANSFER LEARNING STRUGGLES WHEN:

1. MASSIVE DOMAIN SHIFT

- Example: Natural images → Medical microscopy



2. FUNDAMENTALLY DIFFERENT TASKS

- Example: Classification model → Segmentation task



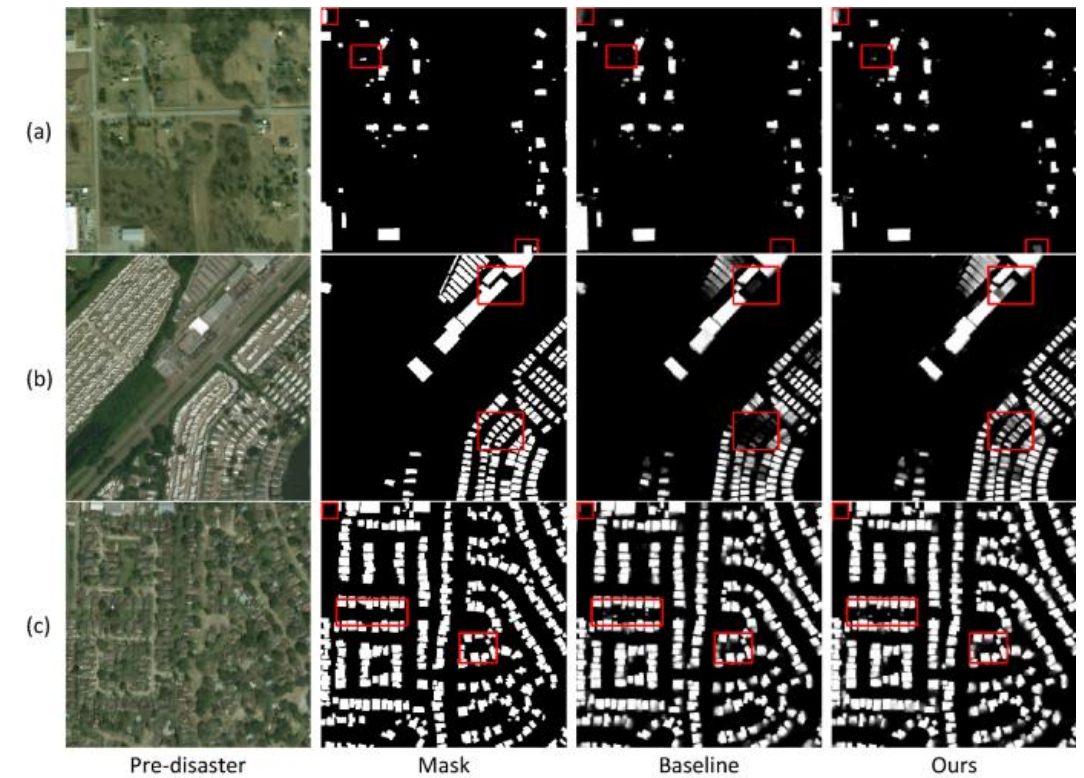
3. EXTREMELY LIMITED DATA

- Example: <50 labeled images

4. TARGET DOMAIN HAS UNIQUE FEATURES

- Example: Underwater imaging (unique distortions, colors)

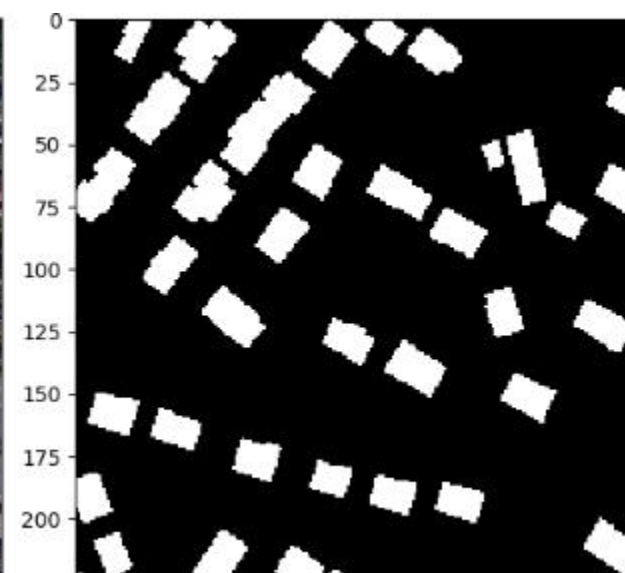
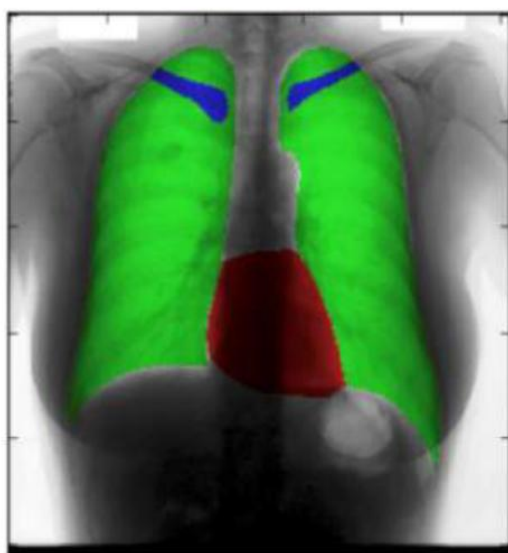
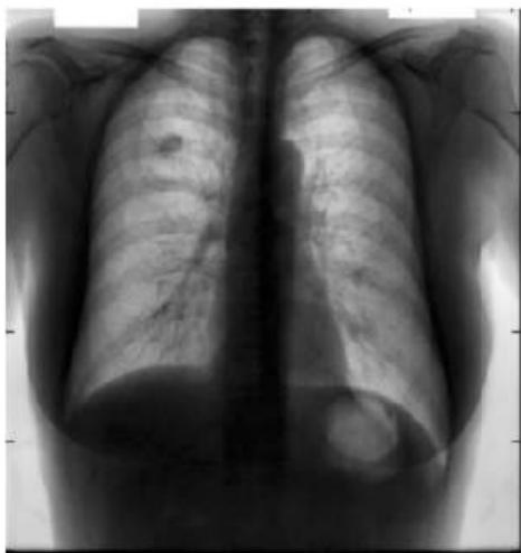
RULE OF THUMB: If a human looking at source and target images sees obvious similarities (objects, scenes, lighting), TL will likely work well.



Introduction to U-Net

MOTIVATION FOR U-NET

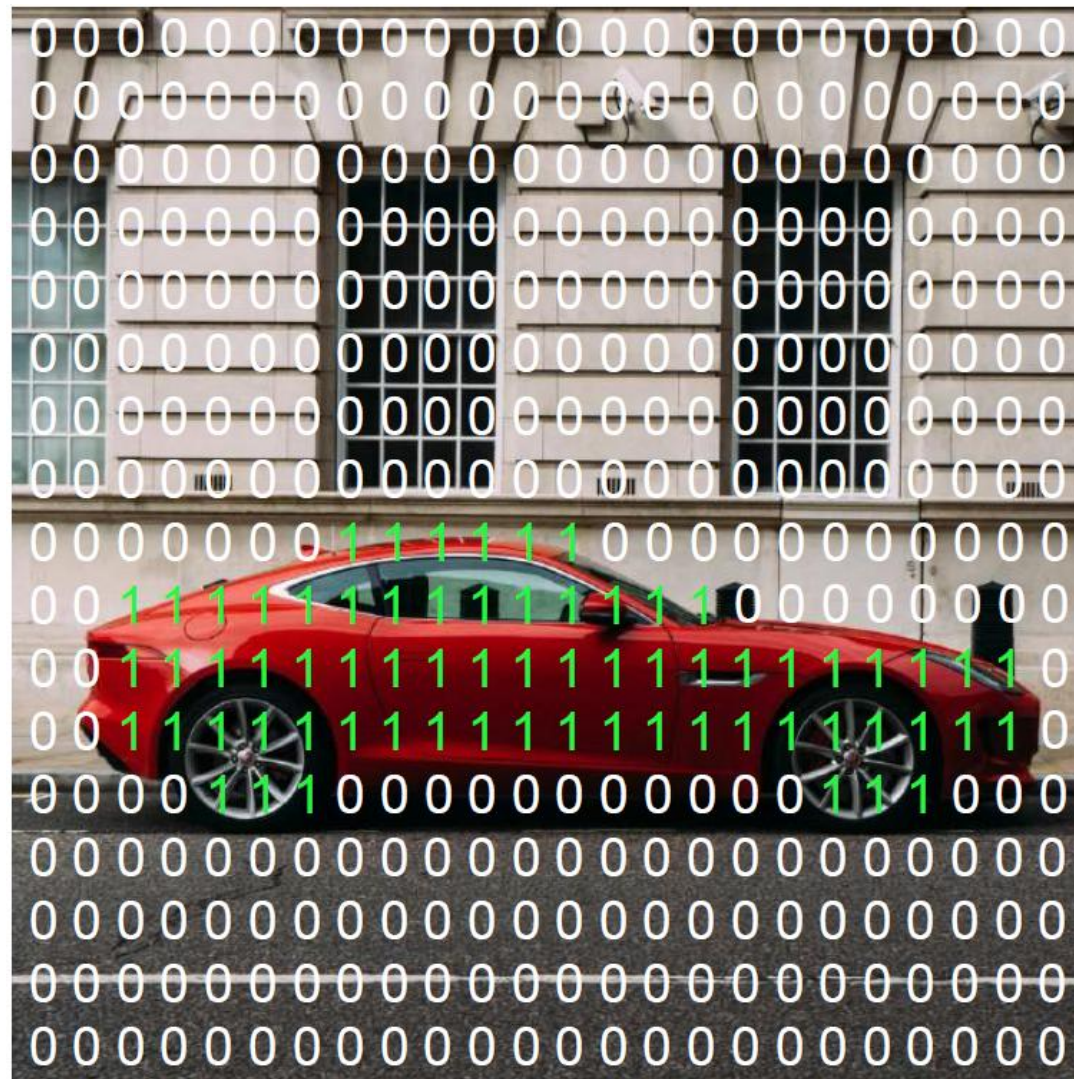
Why do we need architectures like U-Net for segmentation tasks?



Chest X-Ray

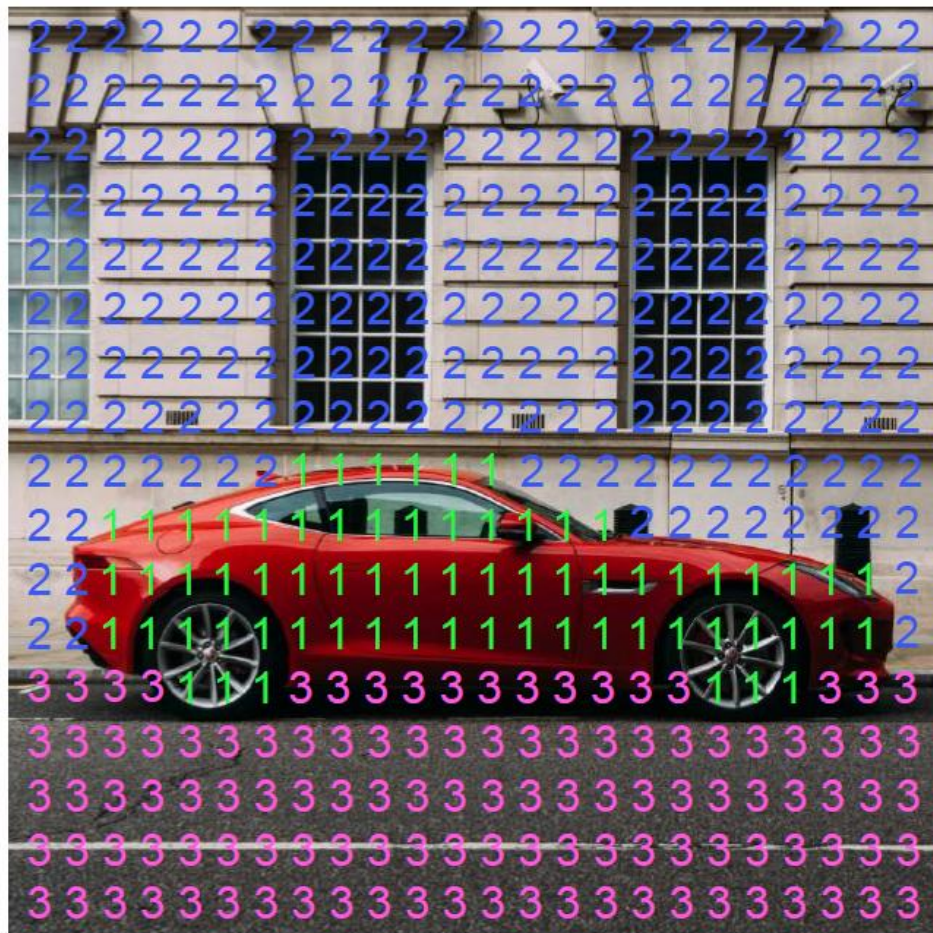
PER-PIXEL CLASS LABELS

What if we need to know exactly which pixels belong to an object? That's where segmentation comes in



1. Car
0. Not Car

PER-PIXEL CLASS LABELS



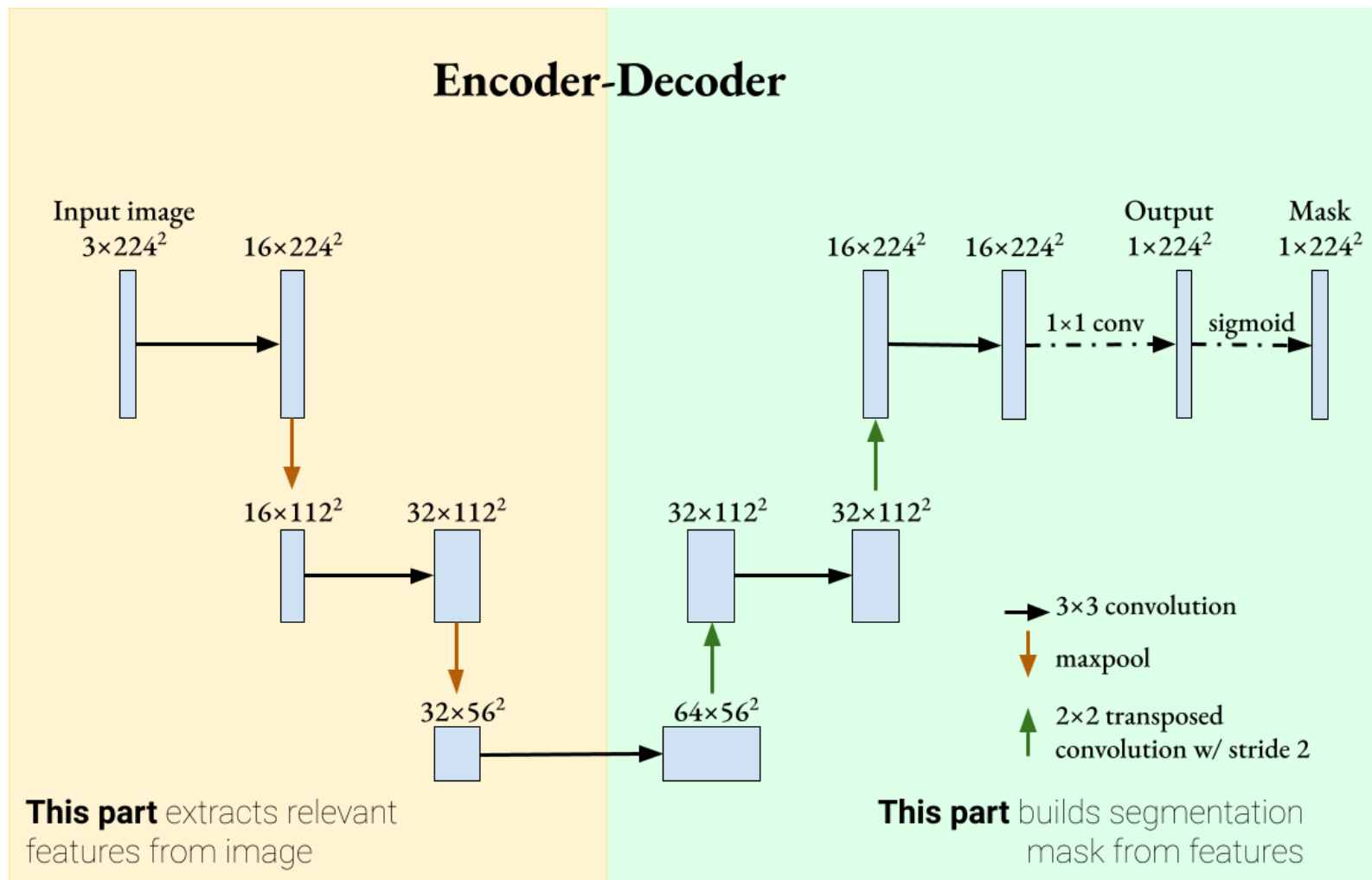
1. Car
2. Building
3. Road



Segmentation Map

❑ ORIGIN & PURPOSE

- Developed: 2015, University of Freiburg-
- Original use: Biomedical image segmentation
- Key innovation: Works with very few training images
- Perfect for: Dense pixel-wise predictions



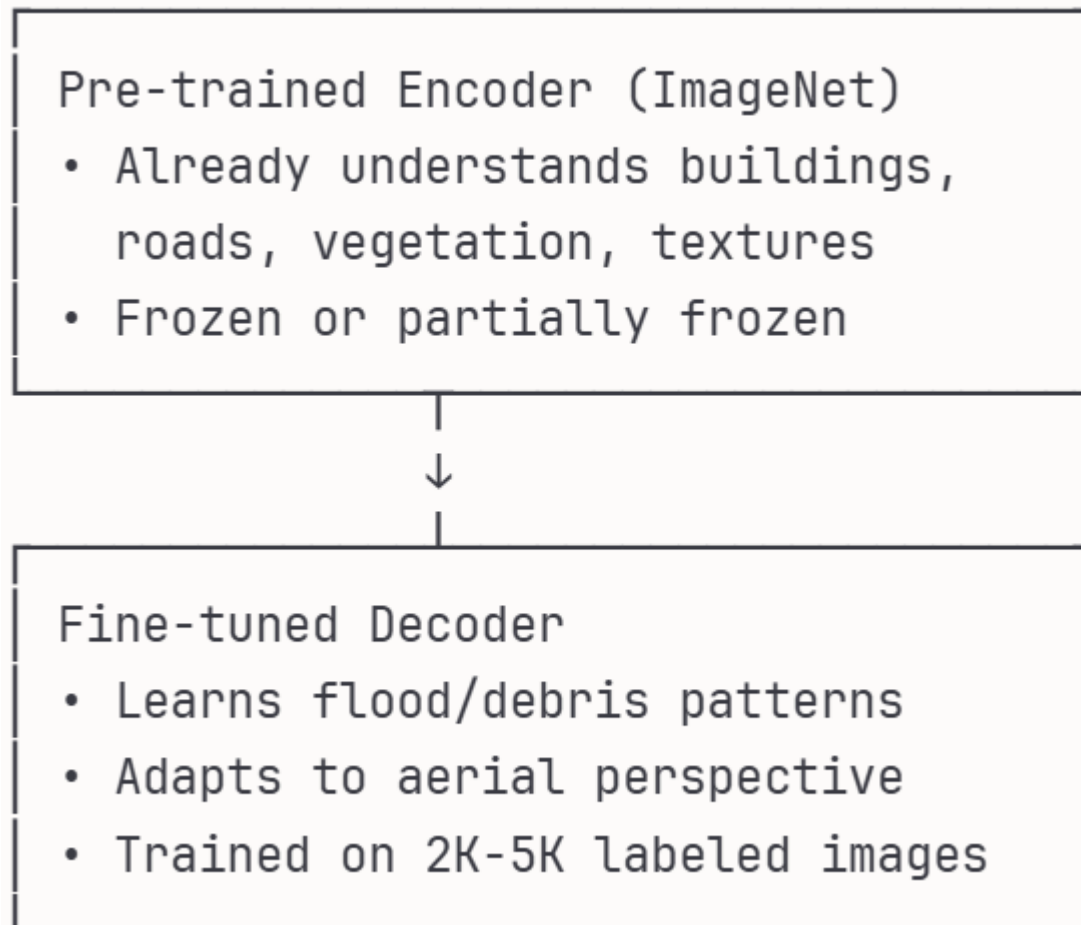
USE CASE 1: FLOOD EXTENT MAPPING

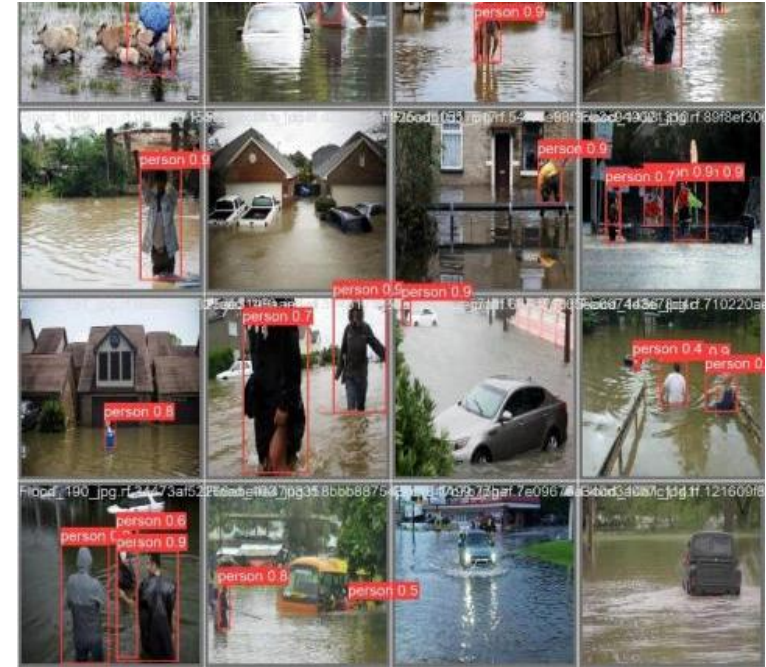
- ❖ Input: Aerial/satellite imagery of affected area
- ❖ Output: Pixel-wise flood mask (flooded vs. non-flooded)
- ❖ OPERATIONAL VALUE
 - Rapid damage assessment (in minutes)
 - Precise area calculations (square meters of flooding)
 - Temporal analysis (flood progression over time)

USE CASE 2: DEBRIS FIELD SEGMENTATION

- ❖ Input: Post-disaster aerial imagery
- ❖ Output: Debris vs. intact structure vs. clear area
- ❖ OPERATIONAL VALUE
 - Structural damage quantification
 - Debris removal prioritization

TRANSFER LEARNING STRATEGY





Introduction to YOLO

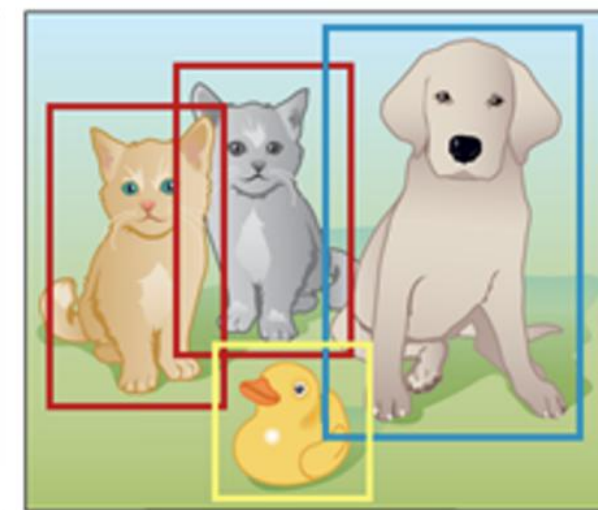
YOLO MODEL FOR OBJECT DETECTION

- ❖ Object detection is the **problem of both locating AND classifying objects**
- ❖ YOLO is a real-time object detection algorithm introduced by Joseph Redmon et al. in 2016.
- ❖ Goal of YOLO algorithm is to do object detection both fast AND with high accuracy.
- ❖ Key Idea is **to treat object detection as a single regression problem** from image pixels to bounding box coordinates and class probabilities.
- ❖ Why YOLO? Extremely fast and efficient compared to traditional region-based methods (e.g., R-CNN).

Object Detection vs Classification
Image classification Object detection
(classification and localization)

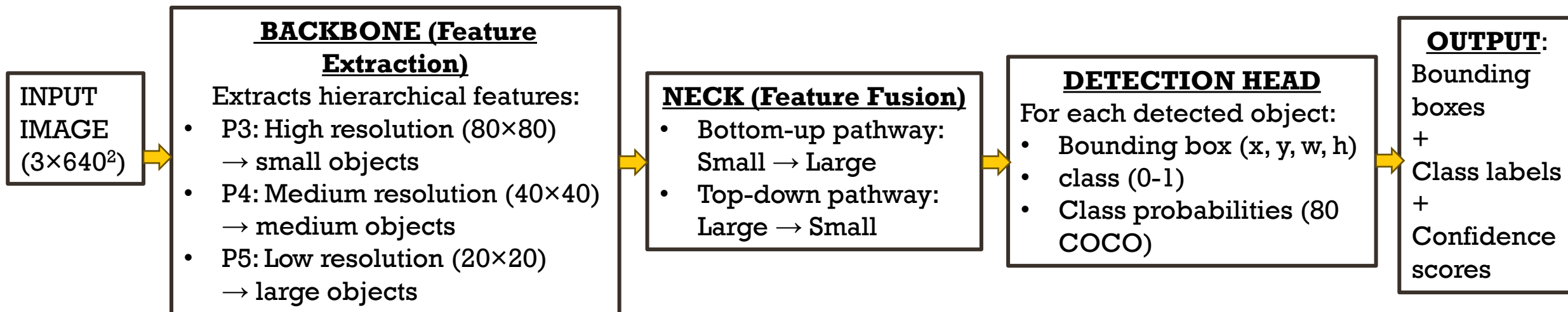


Cat



Cat, Cat, Duck, Dog

YOLO V8 ARCHITECTURE FOR DETECTION



Model	Params	FLOPs	Speed	mAP50
YOLOv8n	3.2M	8.7G	640 FPS	37.3%
YOLOv8s	11.2M	28.6G	420 FPS	44.9%
YOLOv8m	25.9M	78.9G	180 FPS	50.2%
YOLOv8l	43.7M	165G	90 FPS	52.9%
YOLOv8x	68.2M	258G	60 FPS	53.9%



YOLO FOR DISASTER RESPONSE



OPTiSIM
tion Lab

- ❖ Input: Aerial imagery from drones/helicopters
- ❖ Output: Bounding boxes around people + confidence scores + locations.

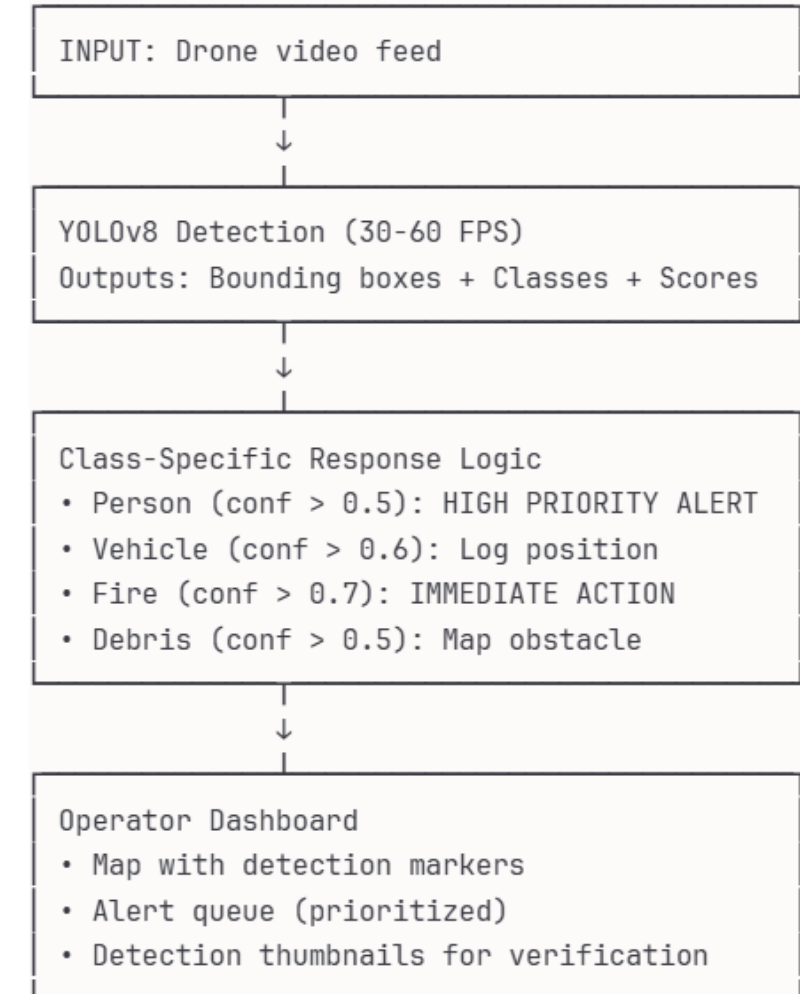
USE CASE 1: SURVIVOR DETECTION

- Drone flies search pattern over affected area
- YOLO processes video feed in real-time (30-60 FPS)
- Operators verify detections, dispatch rescue teams

USE CASE 2: VEHICLE DETECTION & TRACKING

- Evacuation monitoring: Count vehicles leaving area
- Route assessment: Identify blocked roads/bridges

MULTI-CLASS DETECTION PIPELINE



Load Pretrained Model

- Choose YOLO variant (e.g., YOLOv5, YOLOv8) pretrained on COCO.

Freeze Backbone Layers

- Keep feature extraction layers fixed to retain learned representations.

Replace Detection Head

- Adapt final layers for new classes in your dataset.

Fine-Tune

- Train detection head and optionally unfreeze some backbone layers for better adaptation.

PRE-TRAINED YOLO KNOWLEDGE (COCO Dataset)

- 80 object classes including:
 - **Person** (64K examples) → Direct transfer to survivor detection
 - **Vehicles** (43K examples) → Car, truck, bus detection
 - **Furniture** → Context for building interiors
- Diverse environments: Indoor, outdoor, urban, rural
- Various scales: Tiny to large objects
- lighting, weather variations
- 200K total labeled instances

WHAT TRANSFERS WELL

- Low-level features: Edges, corners, textures
- Mid-level features: Object parts, shapes, patterns
- High-level features: Generic object understanding



RESOURCES FOR FURTHER LEARNING

- Zhao, Z., Alzubaidi, L., Zhang, J., Duan, Y., & Gu, Y. (2024). A comparison review of transfer learning and self-supervised learning: Definitions, applications, advantages and limitations. *Expert Systems with Applications*, 242, 122807.
- Ma, Y., Chen, S., Ermon, S., & Lobell, D. B. (2024). Transfer learning in environmental remote sensing. *Remote Sensing of Environment*, 301, 113924.
- Bhardwaj, D., Nagabhooshanam, N., Singh, A., Selvalakshmi, B., Angadi, S., Shargunam, S., ... & Rajaram, A. (2025). Enhanced satellite imagery analysis for post-disaster building damage assessment using integrated ResNet-U-Net model. *Multimedia Tools and Applications*, 84(5), 2689-2714.
- Harish, K., Abinav, H., & Kumar, D. (2025, May). Deep Learning-Driven Disaster and Aerial Image Segmentation using U-Net and Multi-U-Net Architectures. In *International Conference on Sustainability Innovation in Computing and Engineering (ICSICE 2024)* (pp. 401-409). Atlantis Press.
- Sri, T. A., & Kapu, P. (2025, September). Optimizing disaster recovery processes using satellite imaging and U-net architecture. In *AIP Conference Proceedings* (Vol. 3342, No. 1, p. 030004). AIP Publishing LLC.
- Piran, M. J., Wang, X., Kim, H. J., & Kwon, H. H. (2024). Precipitation nowcasting using transformer-based generative models and transfer learning for improved disaster preparedness. *International Journal of Applied Earth Observation and Geoinformation*, 132, 103962.
- Azad, R., Aghdam, E. K., Rauland, A., Jia, Y., Avval, A. H., Bozorgpour, A., ... & Merhof, D. (2024). Medical image segmentation review: The success of u-net. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Bany Abdelnabi, A. A., & Rabadi, G. (2024). Human detection from unmanned aerial vehicles' images for search and rescue missions: a state-of-the-art review. *IEEE Access*.
- Bany Abdelnabi, A. A., Soykan, B., & Rabadi, G. (2024, November). Deep Learning for Human Detection from Aerial Images in Search and Rescue Missions. In *2024 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)* (pp. 1-6). IEEE.
- Jiang, P., Ergu, D., Liu, F., Cai, Y., & Ma, B. (2022). A Review of Yolo algorithm developments. *Procedia computer science*, 199, 1066-1073.
- Vijayakumar, A., & Vairavasundaram, S. (2024). Yolo-based object detection models: A review and its applications. *Multimedia Tools and Applications*, 83(35), 83535-83574.



Thank you!



Ahmad.BanyAbdelnabi@ucf.edu